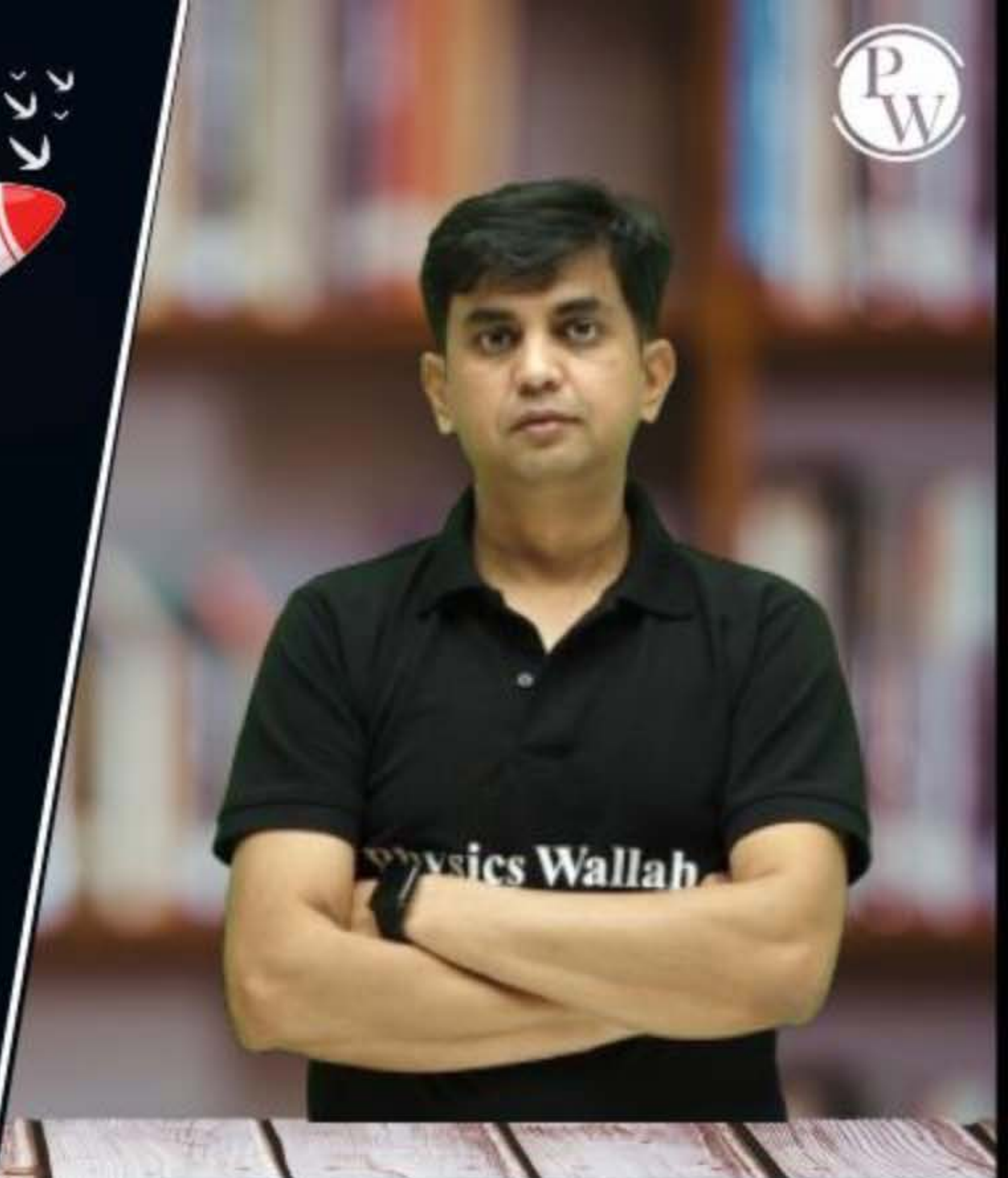
CS & IT ENGINEERING

Computer Network

Routing



Lecture No. – 02

By - Abhishek Sir

Recap of Previous Lecture

Topic Types of Routing

Topics to be Covered

Topic Link State Routing

Topic Distance Vector Routing

Topic : Static Routing

→ Non-adaptive Routing

→ Types of Static Routing :

1. Shortest Path Algorithm
2. Flooding

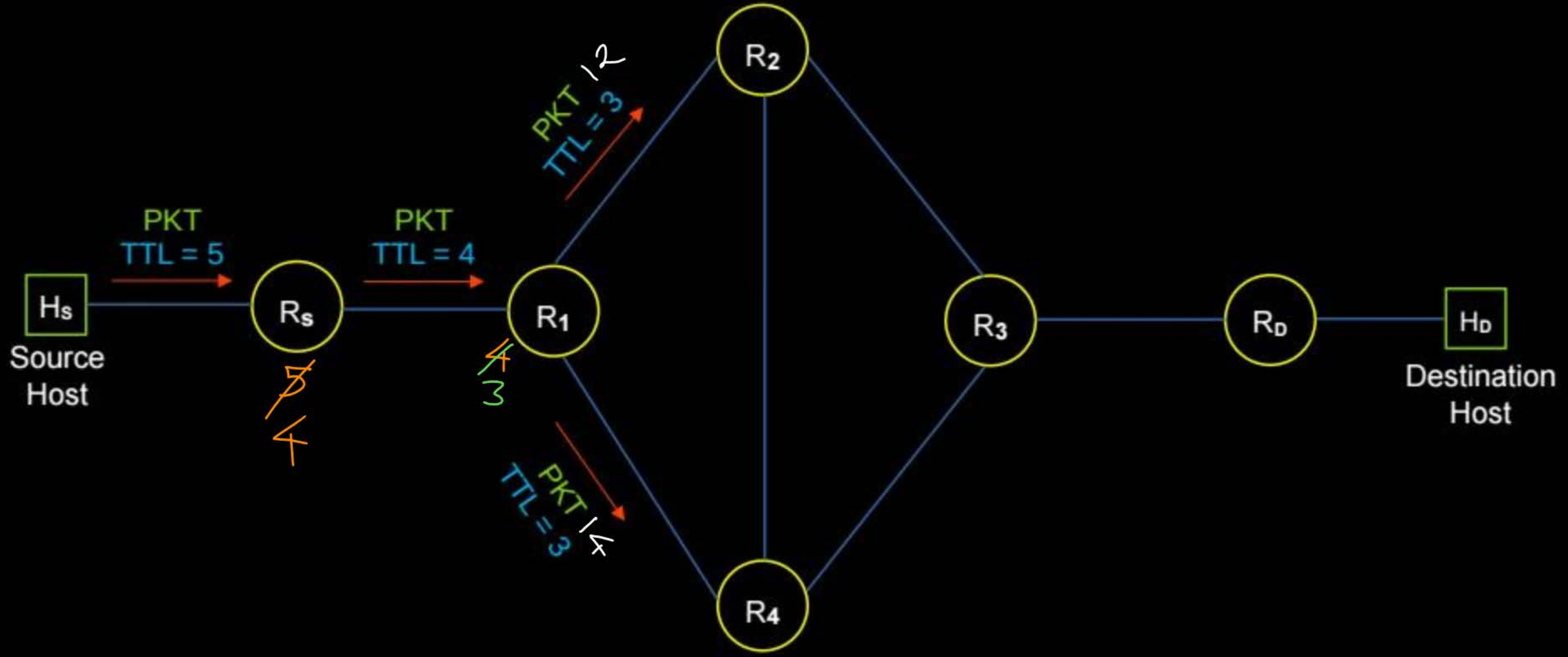
Topic : Flooding

→ Router always broadcast arrived packet
[sent to all the outgoing links except incoming link]

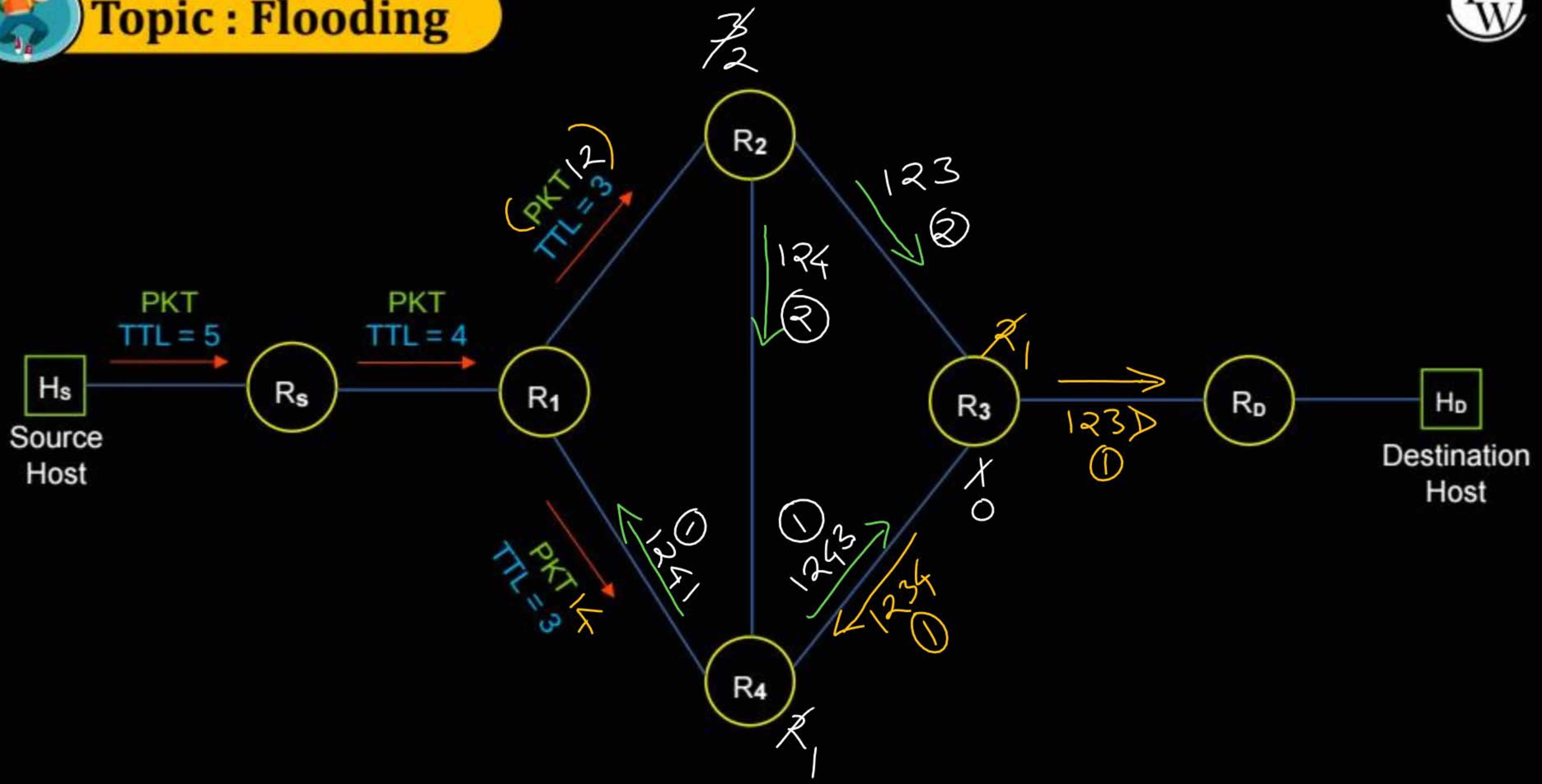
→ Vast number of packets in the network Disadv.

→ The packet which follow shortest path deliver first

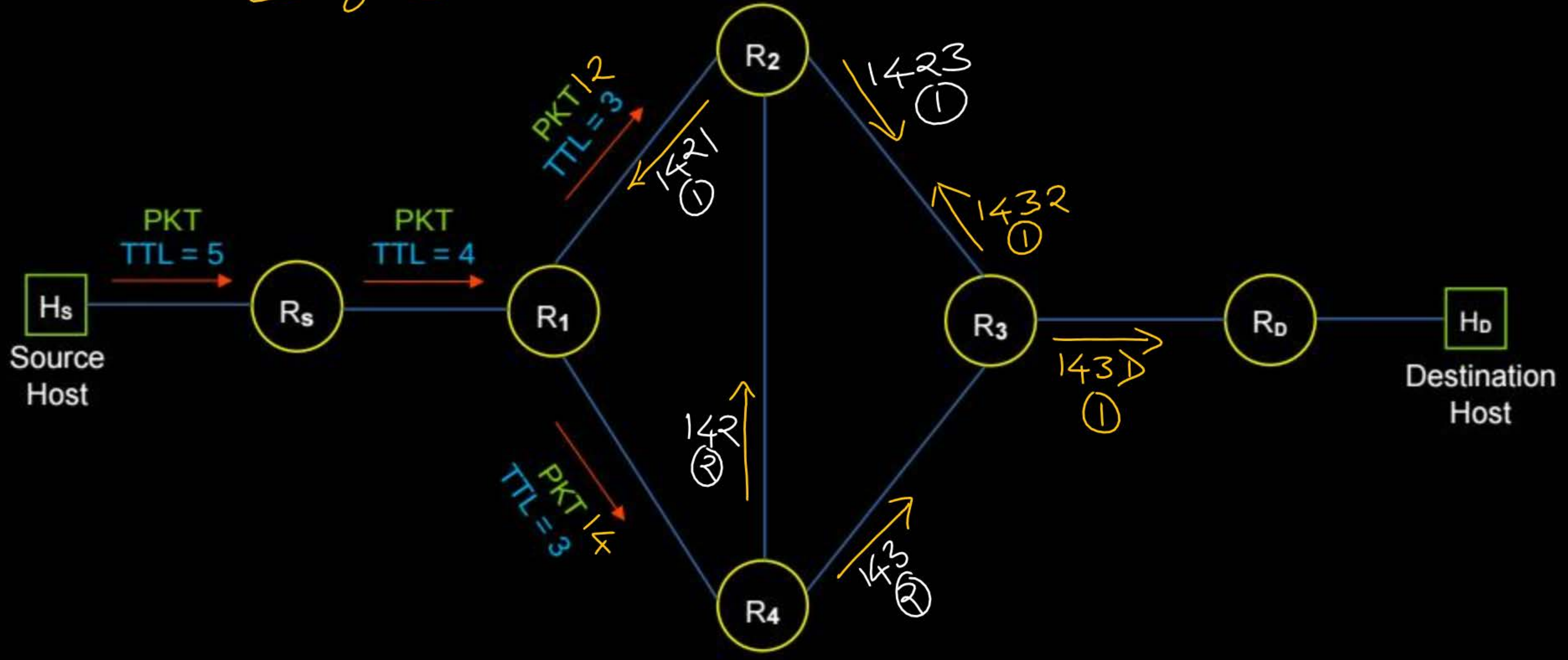
Topic : Flooding



Topic : Flooding



Topic : Flooding



Topic : Dynamic Routing

→ Adaptive Routing

→ Types of Dynamic Routing :

1. Dijkstra's Link State Routing Algorithm
2. Bellman Ford Distance Vector Routing Algorithm

[MCQ] IIT-R [GATE-2025] [1 Mark]

#Q. Consider the routing protocols given in List I and the names given in List II:

List I	List II
(i) Distance vector routing →	(a) Bellman-Ford
(ii) Link state routing →	(b) Dijkstra

For matching of items in List I with those in List II, which ONE of the following options is CORRECT?

(A) (i) – (a) and (ii) – (b)

(B) (i) – (a) and (ii) – (a)

(C) (i) – (b) and (ii) – (a)

(D) (i) – (b) and (ii) – (b)

$\boxed{\text{Ans: A}}$

Topic : Routing Protocols

→ To exchange routing information between routers

→ Two categories :

1. Interior Gateway Protocol [IGP]

2. Exterior Gateway Protocol [EGP]

Topic : Interior Gateway Protocol

→ Intra-domain routing

→ within an autonomous system
(a network or group of networks)

→ Routing Protocols :

1.1 Routing Information Protocol (RIP)

→ Distance Vector Routing

RIP - DV - Bellman Ford

1.2 Open Shortest Path First (OSPF)

→ Link State Routing

OSPF - LS - Dijikstra

[MCQ] IIT-KGP [GATE-2014] [1 Mark]

#Q. Which one of the following is TRUE about (interior Gateway routing protocols) - Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) ?

RIP-DV OSPF-LS

A. RIP uses distance vector routing and OSPF uses link state routing

B. OSPF uses distance vector routing and RIP uses link state routing

C. Both RIP and OSPF use link state routing

D. Both RIP and OSPF use distance vector routing

$\boxed{\text{Ans: A}}$

Topic : Exterior Gateway Protocol

→ Inter-domain routing

→ Between different autonomous system on the internet

→ Routing Protocols :

2.1 Border Gateway Protocol (BGP)

→ Path Vector Routing

Topic : Link State Routing

→ Centralized Routing Algorithm

[determine least-cost path using complete picture of the network topology]

→ Link State Routing is divided into three steps :

1. Maintain Link State information
2. Link State broadcast
3. Share updated Link State information

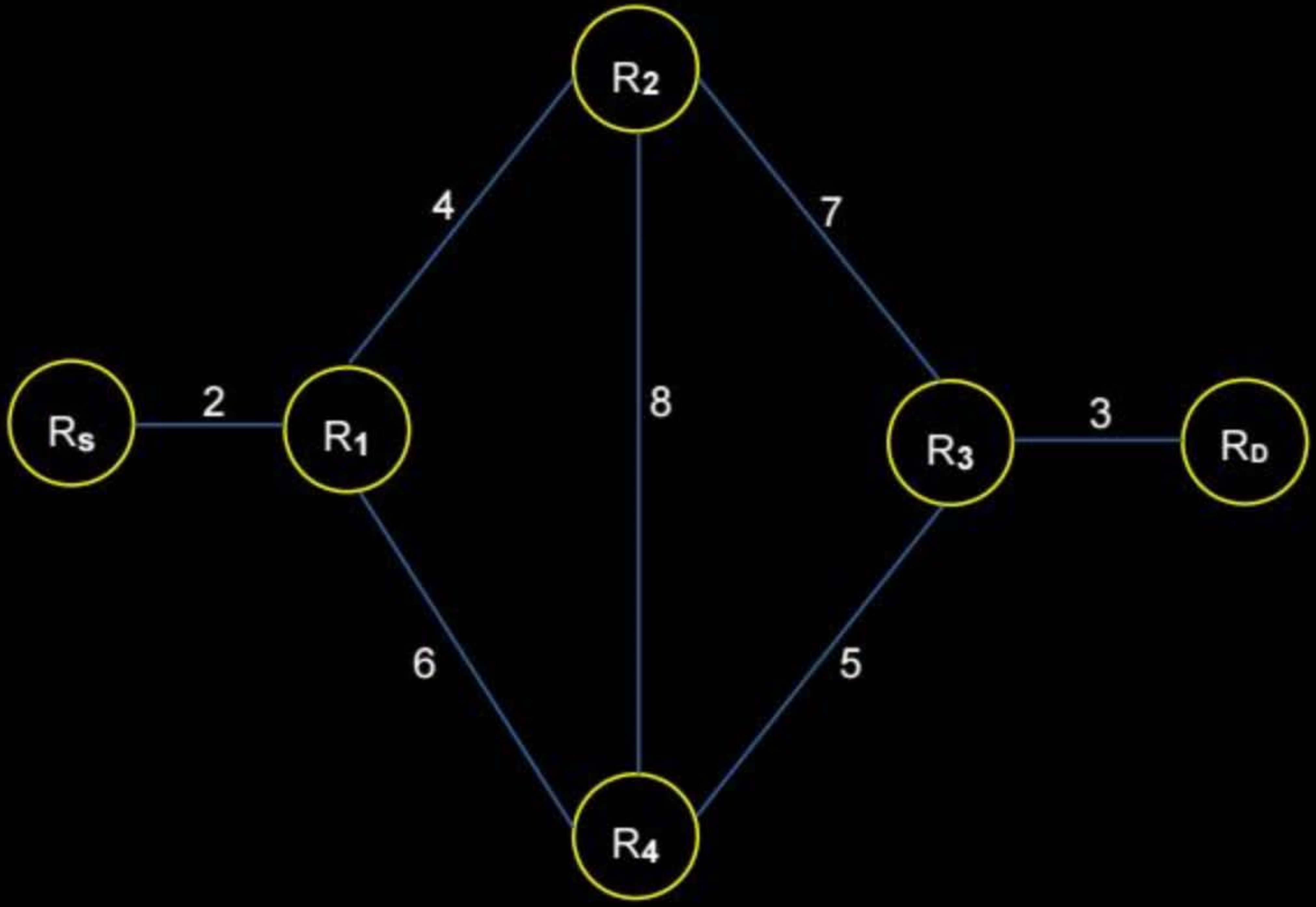
Topic : Link State Routing

1. Maintain Link State information

→ Each router maintain separate "Link State" information

→ Maintain information about adjacent link only
[Adjacent (neighbor) routers only]

Topic : Link State Routing



Topic : Link State Routing

R_S Link State Info	
R_1	2

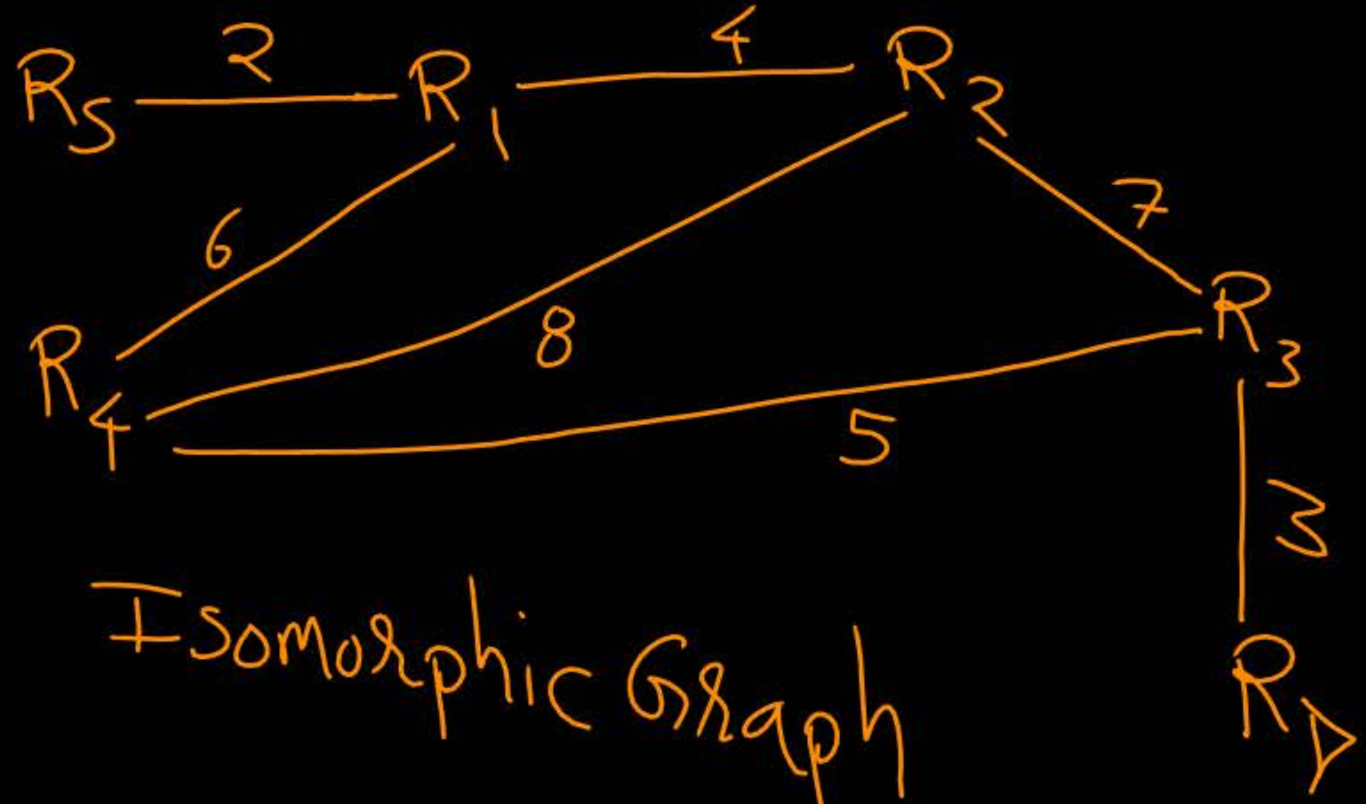
R_D Link State Info	
R_3	3

R_1 Link State Info	
R_S	2
R_2	4
R_4	6

R_4 Link State Info	
R_1	6
R_2	8
R_3	5

R_2 Link State Info	
R_1	4
R_3	7
R_4	8

R_3 Link State Info	
R_D	3
R_2	7
R_4	5



Isomorphic Graph

Topic : Link State Routing

2. Link State broadcast

→ Each router flood (broadcast) its "Link State" information
[To all other routers in the network]

→ Each router have "Link State" information of all other routers
[Complete information about the network topology]

→ Each router construct "Adjacency Matrix" or "Adjacency List"

→ Each router execute "Dijkstra's Algorithm" locally ⟹ Forwarding Table
[To find optimal path to all other routers]

Topic : Link State Routing

Graph
[Adjacency Matrix or List]

Source

Dijkstra Algorithm

Shortest path to all other nodes

Topic : Link State Routing

3. Share updated Link State information ⟹ Asynchronous Mode
⟹ Synchronous Mode

→ Whenever changes occur in the network topology,
[Relevant routers update their "Link State" information accordingly]

→ Only those routers broadcast their updated "Link State" information
[To all other routers in the network]

→ Whenever a router receive any updated "Link State" information,
it construct new "Adjacency Matrix / List" accordingly and
re-execute "Dijkstra's Algorithm" locally

Topic : Dijkstra's Algorithm

Dijkstra's Algorithm time complexity

→ Number of nodes = n

→ Total n iteration

→ Total number of comparisons = $O(n^2)$

Topic : Link State Routing

Message Complexity :

→ Number of nodes = n

→ Each router must broadcast its Link State information to all other router

→ Each router's message crosses O(n) links

→ Overall message complexity = $\boxed{O(n^2)}$

Topic : Link State Routing

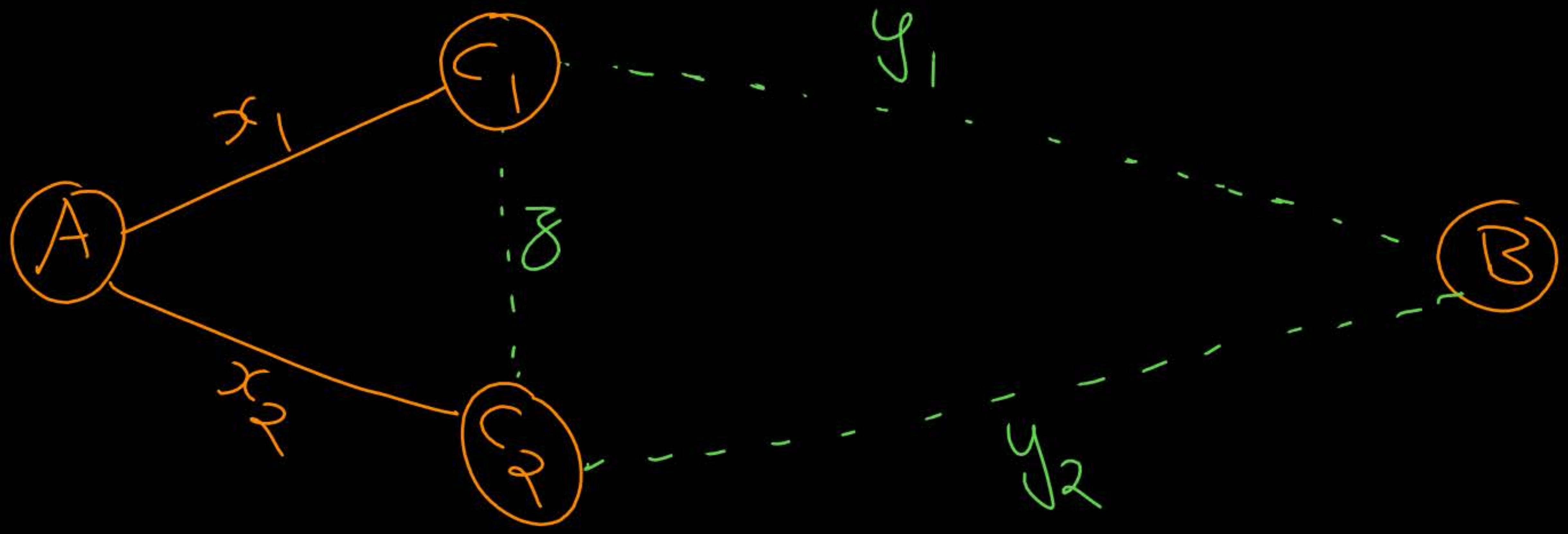
→ High number of packets per network (Disadv.)

→ Adapt network topology very fast (Adv.)

Topic : Distance Vector Routing

→ Based on Bellman-Ford equation
[Dynamic programming]

$$D_A(B) = \min[(x_1+y_1), (x_2+y_2)]$$



Topic : Distance Vector Routing

→ Bellman-Ford equation

Let $D_x(Y)$: Cost of the least-cost path from X to Y

$$D_x(y) = \min_V \{ C(X, V) + D_V(Y) \}$$

→ Minimum taken over all neighbors V of X

$C(X, V)$: Direct cost of link from X to V

$D_V(Y)$: V's estimated least-cost path cost to Y

Topic : Distance Vector Routing

→ Each router maintain separate "Distance Vector" estimate [Best known minimum distance to all other routers]

→ Each router sends their own "Distance Vector" estimate to their neighbor routers only

→ When a router receives new "Distance Vector" estimate from any neighbor, It update its own "Distance Vector" using Bellman-Ford equation

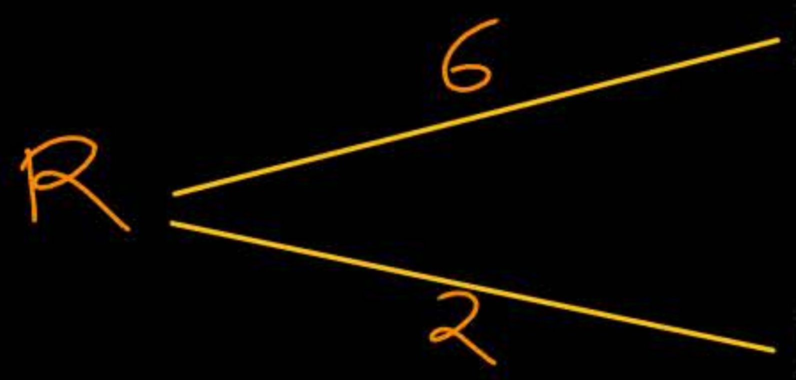
#Q. Consider a network of five routers [P, Q, R, S and T.] Router P and T are direct neighbour of router R with distance 6 and 2 respectively. Consider following are the distance vector of router P and T then calculate distance vector of router R ?

Router P :

P	Q	R	S	T
0	1	5	5	3

Router T :

P	Q	R	S	T
3	4	2	8	0



Router P :

P	Q	R	S	T
0	1	5	5	3

Router T :

P	Q	R	S	T
3	4	2	8	0

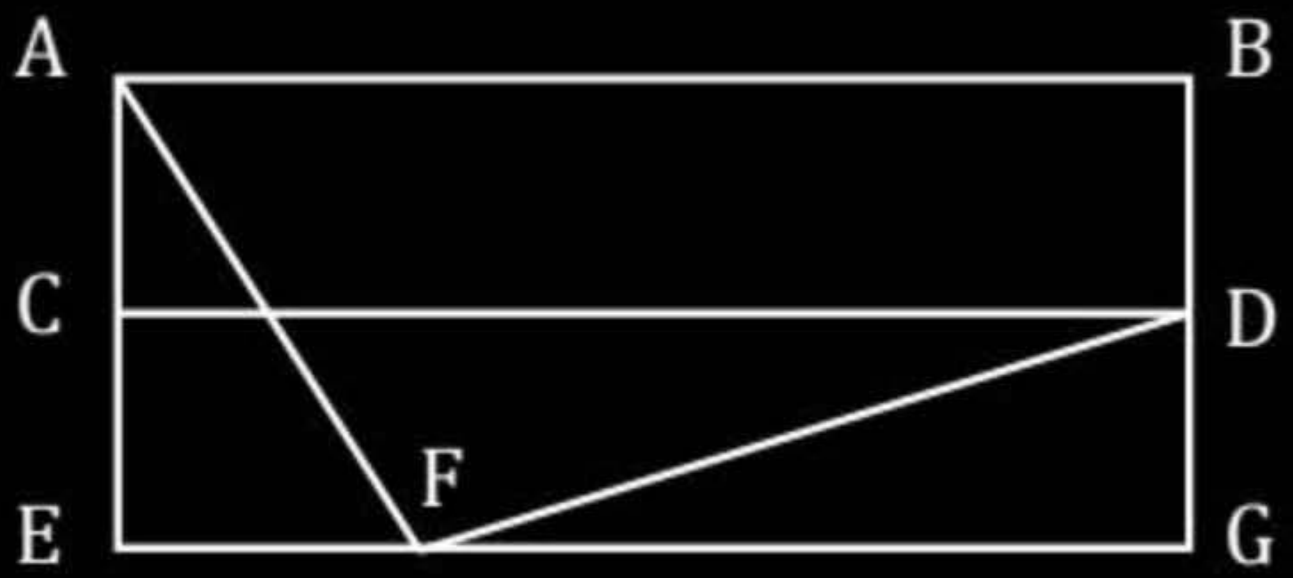
Router R :

Node	P	Q	R	S	T
Cost	5	6	0	10	2
Via	T	T	R	T	T

[MCQ] IIT-K, H.W [GATE-IT-2007] [2 Mark]

#Q. For the network given in the figure below, the routing tables of the four nodes A, E, D and G are shown. Suppose that F has estimated its delay to its neighbors, A, E, D and G as 8, 10, 12 and 6 msecs respectively and updates its routing table using distance vector routing technique.

Routing Table of A		Routing Table of D		Routing Table of E		Routing Table of G	
A	0	A	20	A	24	A	21
B	40	B	8	B	27	B	24
C	14	C	30	C	7	C	22
D	17	D	0	D	20	D	19
E	21	E	14	E	0	E	22
F	9	F	7	F	11	F	10
G	24	G	22	G	22	G	0



A

A	8
B	20
C	17
D	12
E	10
F	0
G	6

B

A	21
B	8
C	7
D	19
E	14
F	0
G	22

C

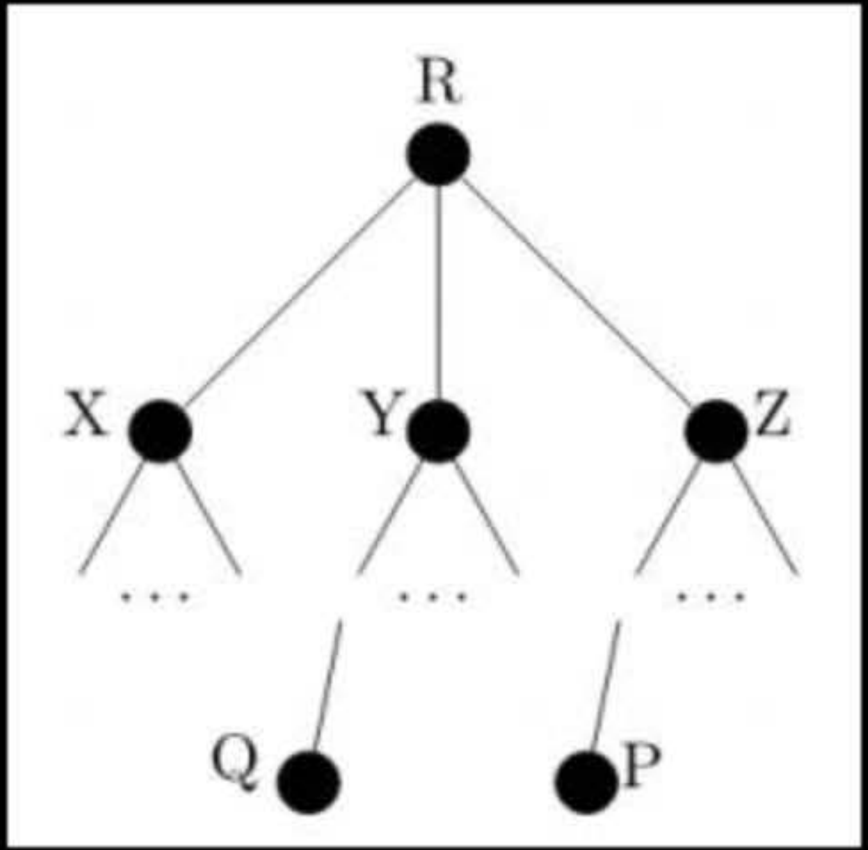
A	8
B	20
C	17
D	12
E	10
F	16
G	6

D

A	8
B	8
C	7
D	12
E	10
F	0
G	6

[MSQ] IIT-B, H.W. [GATE-2021][2 Mark]

#Q. Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is shown below.



The objective is to find the shortest-cost path from the router R to routers P and Q. Assume that R does not initially know the shortest routes to P and Q. Assume that R has three neighbouring routers denoted as X, Y and Z.

During one iteration, R measures its distance to its neighbours X, Y, and Z as 3, 2 and 5, respectively. Router R gets routing vectors from its neighbours that indicate that the distance to router P from routers X, Y and Z are 7, 6 and 5, respectively. The routing vector also indicates that the distance to router Q from routers X, Y and Z are 4, 6 and 8 respectively. Which of the following statement(s) is/are correct with respect to the new routing table of R, after updation during this iteration?

(A) The distance from R to P will be stored as 10

(B) The distance from R to Q will be stored as 7

(C) The next hop router for a packet from R to P is Y

(D) The next hop router for a packet from R to Q is Z

2 mins Summary

Topic Link State Routing

Topic Distance Vector Routing

THANK - YOU

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